

RA8P1 E-READER

Processor schematic development
Power networks are being implemented from RA8P1-specific references.
This revision is not a completed schematic or a fabrication release.

RA8P1 IO allocation



File: mcu_interfaces.kicad_sch

Global supplies: +3V3_MCU and GND.
Core regulator output MCU_VCORE stays local to the processor power sheet.

Remaining subsystem circuits follow epic #821 and issues #824-#834:
radio, memory/storage, USB/battery, system rails, display controller,
panel power, touch, warm/cool lighting, buttons and sensors.
No empty subsystem sheets are represented as completed circuits.

Footprint and PCB qualification are deferred by scope.

RA8P1 internal core regulator



File: mcu_core_power.kicad_sch

RA8P1 IO and analog supplies



File: mcu_supply_power.kicad_sch

RA8P1 clocks, reset and debug



File: mcu_clocks_debug.kicad_sch

Sheet: / File: ereader_rev1.kicad_sch		
Title: RA8P1 e-reader - schematic index		
Size: A3	Date: 2026-09-05	Rev: 0.3 DRAFT
KiCad E.D.A. 10.0.5	Id: 1/5	

DRAFT: Existing processor units retained. Signal allocation and clocks/debug wiring are not yet complete.

U1B
R7KA8P1KFLCAC#UC0

GPIO P0 / P1			
U13	P000	P100	U6
T13	P001	P101	R5
U14	P002	P102	P5
R12	P003	P103	R4
T11	P004	P104	M6
R11	P005	P105	N7
U12	P006	P106	N6
T12	P007	P107	N5
U11	P008	P108	D8
P11	P009	P109	D7
P10	P010	P110	C8
N10	P011	P111	E8
P9	P012	P112	A4
N9	P013	P113	A2
T8	P014	P114	B3
U8	P015	P115	A3

U1C
R7KA8P1KFLCAC#UC0

GPIO P2 / P3			
C5	P200	P300	B5
B15	P206	P301	C4
A17	P207	P302	A5
		P303	B6
		P304	D10
		P305	C10
		P306	D11
		P307	C11
		P308	C9
		P309	A12
		P310	E10
		P311	B12
		P312	C13

U1D
R7KA8P1KFLCAC#UC0

GPIO P4 / P5			
P17	P400	P500	U7
N17	P401	P501	R8
L14	P402	P502	T7
H13	P403	P503	H2
J13	P404	P504	H1
G12	P405	P505	H3
F14	P406	P506	J1
R17	P407	P507	J2
M15	P408	P508	J3
R16	P409	P509	J4
K13	P410	P510	K3
M14	P411	P511	U15
M12	P412	P512	P13
P14	P413	P513	R13
L13	P414	P514	R14
R15	P415	P515	P15

U1E
R7KA8P1KFLCAC#UC0

GPIO P6 / P7			
P3	P600	P700	F12
P4	P601	P701	F15
P2	P602	P702	F13
P1	P603	P703	G14
N2	P604	P704	G13
N1	P605	P705	F17
N3	P606	P706	E17
M2	P607	P707	F16
K2	P608	P708	N12
A1	P609	P709	P16
D3	P610	P710	M13
D2	P611	P711	N15
E4	P612	P712	N13
C2	P613	P713	N16
E3	P614	P714	N14
E2	P615	P715	T15

U1F
R7KA8P1KFLCAC#UC0

GPIO P8 / P9			
T6	P800	P902	E9
P6	P801	P903	D9
R6	P802	P904	A16
P7	P803	P905	A14
R7	P804	P906	A13
P12	P805	P907	A15
T14	P806	P908	B13
N11	P807	P909	B14
U5	P808	P910	E11
T5	P809	P911	C12
M9	P810	P912	D12
N8	P811	P913	E12
P8	P812	P914	E5
B1	P813	P915	A6

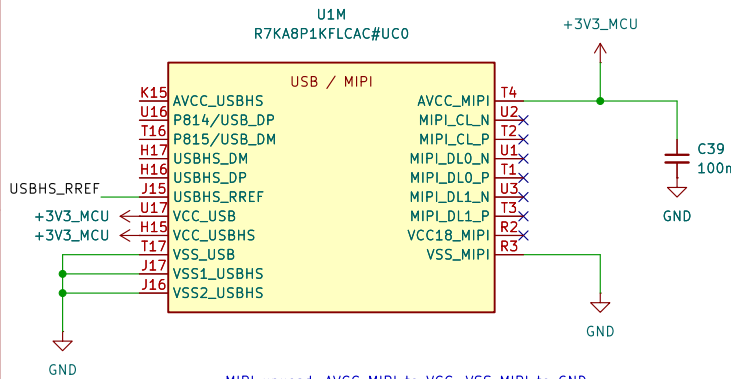
U1G
R7KA8P1KFLCAC#UC0

GPIO PA / PB			
G1	PA00	PB00	E16
H4	PA01	PB01	D17
F1	PA02	PB02	E13
G2	PA03	PB03	D16
D1	PA04	PB04	D13
G3	PA05	PB05	D15
C1	PA06	PB06	E14
G4	PA07	PB07	D14
F3	PA08		
F2	PA09		
B4	PA10		
B2	PA11		
C3	PA12		
D4	PA13		
E1	PA14		
	PA15		

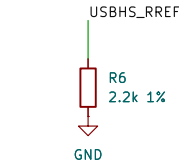
U1H
R7KA8P1KFLCAC#UC0

GPIO PC / PD			
M1	PC00	PD00	K4
M3	PC01	PD01	B16
L2	PC02	PD02	B17
L1	PC03	PD03	C15
L3	PC04	PD04	C14
L5	PC05	PD05	C16
N4	PC06	PD06	C17
K5	PC07	PD07	E15
M4	PC08		
L4	PC09		
M5	PC10		
H5	PC11		
G5	PC12		
J5	PC13		
F5	PC14		
K1	PC15		

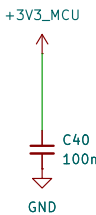
U1M
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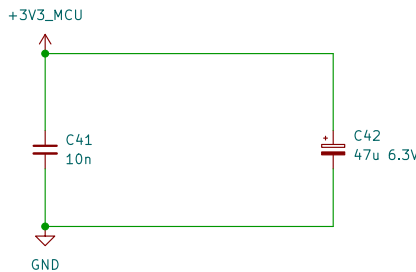
MIPI unused: AVCC_MIPI to VCC; VSS_MIPI to GND.
Leave VCC18_MIPI and all six D-PHY lanes open.
Firmware: MSTPCRC.MSTPC10=1 and MSTPC17=1.
RA8P1 HUM Rev.1.30, section 21.4, p.862.



USB-001: R6 USBHS current-reference resistor
Renesas QDG Table 2, p.7: 2.2 kohm +/-1%.
 $R_{min} = 2200 \times (1 - 0.01) = 2178 \text{ ohm}$
 $R_{max} = 2200 \times (1 + 0.01) = 2222 \text{ ohm}$
Initial tolerance only; not a derived PHY current.
Full basis + Python check: design/usb_interface.md



USB-002: USB full-speed supply / U1,U17, C40.
Renesas QDG Table 1, p.6: 3.3 V with 100 nF bypass.
 $C40 = 100 \times (1 + \sqrt{-0.10}) = 90.110 \text{ nF}$ initial: 50 V X7R.
Nominal DC-bias estimate: 99.45 nF at 3.3 V (PWR-001).
Place C40 close to VCC_USB with a short VSS_USB return.
Full basis + Python check: design/usb_interface.md



USB-003: USB-HS supply / U1,H15, C41, C42
Renesas QDG Table 1 p.6: 10 nF ceramic + 47 uF electrolytic.
 $C41 = 10 \times (1 + \sqrt{-0.10}) = 9.11 \text{ nF}$ initial: 50 V X7R.
TDK nominal bias model: C41(3.3 V) = 10.084 nF (not guaranteed).
 $C42 = 47 \times (1 + \sqrt{-0.20}) = 37.6.56.4 \text{ uF}$ initial: 6.3 V.
Voltage screening: $3.6/6.3 \times 100 = 57.14\%$ rating, not rail sign-off.
C42 positive to 3V3, negative to GND; no reverse voltage.
C41 close to H15; short returns to VSS1/VSS2_USBHS.
Full math + Python / qualification: ../design/usb_interface.md

PWR-001: C39 AVCC_MIPI bypass, 100 nF (QDG Table 2, p.7).
TDK C1608X7R1H104K080AA; also used at C6 and C16-C34.
Initial C = $100 \times (1 + \sqrt{-0.10}) = 90.110 \text{ nF}$; 50 V X7R.
TDK DC-bias samples: 99.5575 nF / 3.15 V; 98.9525 nF / 4 V.
 $C(3.3 \text{ V}) = 99.5575 + (3.3 - 3.15) / 0.85 \times (-0.605) = 99.45 \text{ nF}$.
Interpolated nominal reference, NOT a guaranteed minimum.
Final rail / PDN qualification open; $3.3/50 \times 100 = 6.6\%$ rating.
Full basis + Python check: design/power_decoupling.md

RA8P1 e-reader

Sheet: /RA8P1 IO allocation/
File: mcu_interfaces.kicad_sch

Title: RA8P1 IO allocation

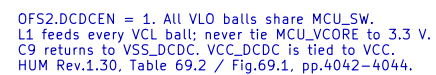
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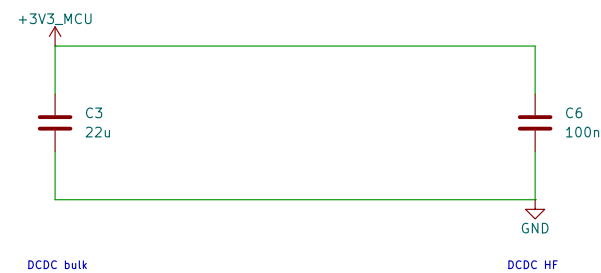
U1K
R7KA8P1KFLCAC#U00

DC/DC

A10 VCC_DCDC VLO A7
A11 VCC_DCDC VLO A8
B10 VCC_DCDC VLO B7
B11 VCC_DCDC VLO B8
A9 VSS_DCDC
B9 VSS_DCDC

+3V3_MCU

GND



L1 DCR <= 100mOhm. Effective capacitance and inductor current rating require final MPN qualification. The 3.3 V source and return will be declared on the system-power sheet; no blanket ERC flags.

